Analysis of Facebook and Google Online Campaign Details

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Tools: DBeaver, PostgreSQL

Indroduction

In this project, I conducted a comprehensive performance analysis using multi-channel (Google and Facebook) digital marketing data. I processed the spending, reach, and revenue metrics in the dataset through daily performance rankings focused on ROMI (Return on Investment), weekly campaign-based value records, and monthly growth trends. Specifically, by measuring the continuity of ad sets through "continuous impression" analysis, I established a structure that combines data analytics capabilities with marketing optimization and decision support processes.

Datasets

1.facebook_ads_basic_daily

	ad_date	campaign_id	adset_id	spend	impressions	reach	clicks	leads	value	
1	2021-11-30	+1TVQ0Ksk5C1R7k9dKSrpg==	kjdNzjgMLmHu9QXPIKaQHQ==	2,686	5,252	4,624	13	[NULL]	4,058	utr
2	2022-04-12	pf4uuqaAAFuHa7gEJnQVKg==	+/ujgC+9ZiLznb2LxP6ihw==	25,154	4,578	16,804	298	[NULL]	31,921	utr
3	2021-06-17	+1TVQ0Ksk5C1R7k9dKSrpg==	QBLf/5REFjL80pmdIrLGCw==	4,242	6,045	2,323	97	0	5,597	[NU
4	2022-01-31	h4hWTQf6icvBNjHfA+zKhw==	MVc+HmWqkmrIUI6Iyjdw4w==	1,693	19,554	3,318	80	2	3,383	utr
5	2022-04-05	pf4uuqaAAFuHa7gEJnQVKg==	+/ujgC+9ZiLznb2LxP6ihw==	45,152	19,189	12,587	244	7	53,175	utr
6	2021-09-11	+1TVQ0Ksk5C1R7k9dKSrpg==	jKVwd+J2wdeRAmaZ/ibNHA==	5,665	12,784	13,012	26	3	6,140	utr
7	2021-08-12	FgcthDK8dH5elxOj5go3CA==	omGufzIW6LYDzVB3CvxDIQ==	15,337	151,554	91,668	1,358	10	26,722	utr
8	2021-10-28	+1TVQ0Ksk5C1R7k9dKSrpg==	jKVwd+J2wdeRAmaZ/ibNHA==	1,351	2,491	1,206	21	0	2,224	utr
9	2021-11-15	FgcthDK8dH5elxOj5go3CA==	LYK3a9gdirWrWX/VHV9Khw==	2,964	5,250	2,771	187	1	3,813	utr
10	2022-06-01	wHoozjerL5GYr6gvvxnn1Q==	ETDduxyyIf03B90ItrGKSg==	4,733	28,678	23,582	166	3	4,646	utr
11	2020-12-01	OL3wfqCd6TPOleVAcp2yew==	aS+Cy6Ne4nDFe/HabBjs0Q==	879	17,468	7,304	463	7	1,354	utr
12	2021-06-12	yaS9WL9giIMlc/ztuJKtdw==	TcH4JjAaOH8sfVIQOYMulg==	786	3,440	3,566	29	1	728	utr
13	2021-02-06	OL3wfqCd6TPOleVAcp2yew==	aS+Cy6Ne4nDFe/HabBjs0Q==	1,226	5,852	8,336	242	2	1,290	utr
14	2021-08-16	0c40WP42N3VsgjIJdjNGXA==	dfx+slA2Xx6Uvb5jbH/Xog==	1,017	15,372	18,629	273	1	900	utr
15	2022-06-06	wHoozjerL5GYr6gvvxnn1Q==	ETDduxyyIf03B90ItrGKSg==	2,963	9,143	25,076	193	3	4,383	utr
16	2021-09-27	+1TVQ0Ksk5C1R7k9dKSrpg==	kjdNzjgMLmHu9QXPIKaQHQ==	5,540	5,105	10,382	40	3	9,511	utr
17	2021-12-02	FgcthDK8dH5elxOj5go3CA==	omGufzIW6LYDzVB3CvxDIQ==	9,532	22,350	11,245	316	4	8,232	utr
18	2021-11-19	FgcthDK8dH5elxOj5go3CA==	XYc+HmWqkmrIUI6Iyjdw4y==	10,946	26,520	73,834	446	9	15,493	utr
19	2021-10-11	FgcthDK8dH5elxOj5go3CA==	LYK3a9gdirWrWX/VHV9Khw==	1,519	1,741	2,420	29	1	2,463	utr
20	2021-08-07	FgcthDK8dH5elxOj5go3CA==	LYK3a9gdirWrWX/VHV9Khw==	8,506	18,070	18,446	188	6	13,158	utr
21	2021-11-13	+1TVQ0Ksk5C1R7k9dKSrpg==	kjdNzjgMLmHu9QXPIKaQHQ==	791	1,985	4,933	14	1	700	utr
22	2021-09-17	+1TVQ0Ksk5C1R7k9dKSrpg==	kjdNzjgMLmHu9QXPIKaQHQ==	8,169	9,032	8,271	38	2	10,570	utr
23	2021-07-22	0NN3/Q28t0sw9rVduSJU5A==	Mw9J34JDdWqKTcf39+5t/g==	4,799	3,797	5,237	120	3	7,309	utr
24	2022-03-21	pf4uuqaAAFuHa7gEJnQVKg==	eyiLXOBiNEWjXJF37q0CeA==	14,342	6,384	9,715	51	3	16,235	utr
25	2021-07-13	+1TVQ0Ksk5C1R7k9dKSrpg==	OWUAmZpRDMbW00J8WQh8UC	3,823	12,838	10,998	33	3	3,268	utr
26	2022-04-28	wHoozjerL5GYr6gvvxnn1Q==	1KXJtup8qnApqauQIn/3ew==	4,755	65,625	23,726	275	4	7,714	utr
27	2021-09-03	0c40WP42N3VsgjIJdjNGXA==	dfx+slA2Xx6Uvb5jbH/Xog==	911	19,262	18,001	318	6	1,005	utr
28	2022-03-25	pf4uuqaAAFuHa7gEJnQVKg==	eyiLXOBiNEWjXJF37q0CeA==	19,775	5,132	11,780	31	1	32,217	utr

2.google_ads_basic_daily

	ad_date	campaign_name	adset_name	spend	impressions	reach	clicks	leads	value
1	2021-11-05	Wholesale	Wide	1,315	4,314	2,769	50	2	1,228
2	2022-05-11	Hobbies	Outdoors	1,175	37,296	17,783	73	5	918
3	2021-11-07	Wholesale	Wide	1,492	901	1,030	34	1	1,026
4	2021-01-31	Discounts	True power	1,269	4,975	3,145	56	1	1,379
5	2021-07-17	New items	Presents	333	234	387	8	0	391
6	2022-05-26	Hobbies	Quick	1,712	14,188	1,861	59	2	1,707
7	2022-04-03	Expansion	Products	27,416	3,260	11,787	99	4	34,284
8	2022-05-24	Hobbies	Abstract	2,662	5,092	16,429	71	1	3,939
9	2022-05-25	Hobbies	Abstract	1,710	12,751	7,163	48	0	1,449
10	2022-10-29	Expansion	Products	49,801	26,075	15,090	81	0	67,070
11	2022-06-01	Hobbies	Specific	3,658	12,384	8,452	41	3	3,546
12	2022-04-07	Expansion	Products	19,282	15,956	17,488	89	3	12,754
13	2021-07-23	Electronics	Narrow	17,467	55,943	59,874	330	5	27,647
14	2021-06-25	Wholesale	Interests	2,516	4,105	5,003	50	1	2,845
15	2021-11-21	Wholesale	Wide	891	873	3,089	14	0	1,334
16	2020-11-12	Discounts	Upgrades	66	180	129	7	0	76
17	2021-08-27	Crazy discounts	Improvement	1,480	3,515	9,056	127	3	2,211
18	2022-06-18	Hobbies	Tinkering	8,757	49,340	15,130	195	3	6,948
19	2022-04-02	Hobbies	Tinkering	1,072	44,479	11,145	330	6	1,389
20	2021-06-25	Promos	Parents	2,522	19,733	10,117	56	1	1,895
21	2021-08-26	Electronics	Narrow	15,246	28,026	48,569	214	12	21,434
22	2021-07-06	Wholesale	Wide	2,591	6,429	1,264	35	1	3,327

3.facebook_adset

	adset_id	adset_name
1	ZW79Y4iAC9tYUZ0rv2yQdA==	Travel
2	Zq23ozPRs6SFhGqG7C28Sw==	Streaming
3	xorZEO1JrGXyHxvyxdv5Eg==	Lifestyle
4	UIqcMN1yaehmJD9QIXFmqg==	Top gear
5	tx5Zo5YqSvAumn3DuI7T6g==	Cringe
6	tuYFxKP7mMhEGWw7NeEPTw==	Reach 3
7	TKWXp/AU/OvmicTsdwJxHg==	Development
8	TcH4JjAaOH8sfVIQOYMuIg==	Presents
9	RARrsTtOviYO5s61ISG9YA==	Reach 4
10	QBLf/5REFjL80pmdIrLGCw==	Starting up
11	OWUAmZpRDMbW00J8WQh8UQ==	Wide
12	omGufzlW6LYDzVB3CvxDIQ==	Broad
13	nVih5YGpAhIH5K82XEn1Ow==	Outdoors
14	NVhnPAcdYyNcN44Qbfexxw==	Upgrades
15	Mw9J34JDdWqKTcf39+5t/g==	Parents
16	XYc+HmWqkmrlUI6lyjdw4y==	Trending new
17	MVc+HmWqkmrlUI6lyjdw4w==	Trending new
18	mMVUJgUJWaZ7irsgOzCOGw==	Unlimited
19	M2bqycLxxMg4Sb1OesrcPg==	Spotlight
20	LYK3a9gdirWrWX/VHV9Khw==	Broad new
21	IQGYwxBbEAXwpd1MiW0sXw==	Research
22	kjdNzjgMLmHu9QXPIKaQHQ==	Clean
23	iKVwd+I2wdeRAmaZ/ibNHA==	Interests

4.facebook_campaign

	campaign_id	campaign_name
1	OL3wfqCd6TPOIeVAcp2yew==	Discounts
2	wZXc8Eos3JfIQ0EbtDy/Hw==	Brand
3	Kek1yMtTa5BOTQQs2/uEPg==	Electronics
4	0NN3/Q28t0sw9rVduSJU5A==	Promos
5	yaS9WL9gilMIc/ztuJKtdw==	New items
6	FgcthDK8dH5elxOj5go3CA==	Lookalike
7	0c40WP42N3VsgjIJdjNGXA==	Crazy discounts
8	+1TVQ0Ksk5C1R7k9dKSrpg==	Wholesale
9	h4hWTQf6icvBNjHfA+zKhw==	Trendy
10	wHoozjerL5GYr6gvvxnn1Q==	Hobbies
11	pf4uuqaAAFuHa7gEJnQVKg==	Expansion

1. Showing the average,maximum, and minimum values of daily spending metrics separately for Google and Facebook

1.1 SQL Query

```
WITH table_union as (
SELECT ad_date, spend, 'facebook' as ad_type
FROM facebook_ads_basic_daily
UNION ALL
SELECT ad_date, spend, 'google' as ad_type
FROM google_ads_basic_daily
)
SELECT
ad_date,
ad_type,
avg(spend) as avg_spend,
max(spend) as max_spend
min(spend) as min_spend
FROM table_union
GROUP BY ad_date, ad_type
ORDER BY ad_date;
```

1.2 Obtained Results

	ad_date	ad_type	avg_spend	max_spend	min_spend
1	2020-11-11	facebook	189	189	189
2	2020-11-12	facebook	1,150	2,266	34
3	2020-11-12	google	199	332	66
4	2020-11-13	google	1,054	1,054	1,054
5	2020-11-13	facebook	318	622	14
6	2020-11-14	google	1	1	1
7	2020-11-14	facebook	727	727	727
8	2020-11-15	facebook	698	698	698
9	2020-11-15	google	682	682	682
10	2020-11-16	google	2,760	2,760	2,760
11	2020-11-16	facebook	4,167	4,167	4,167
12	2020-11-17	facebook	1,261	1,261	1,261
13	2020-11-18	google	372	372	372
14	2020-11-18	facebook	3,330	3,330	3,330
15	2020-11-19	facebook	1,083	1,083	1,083

A suitable result has been obtained to observe how minimum, maximum, and average advertising costs change over time.

2. Finding the top 5 days in terms of total ROMI

2.1 SQL Query

```
WITH romi_list as (
SELECT
ad_date,
'facebook' as ad_type,
ROUND((COALESCE(((((SUM(value)::float - SUM(spend)::float) /
NULLIF(SUM(spend)::float, 0)))*100),0))::numeric,2)  AS ROMI
FROM facebook_ads_basic_daily
GROUP BY ad_date
UNION ALL
SELECT
ad_date,
'google' as ad_type,
ROUND((COALESCE(((((SUM(value)::float - SUM(spend)::float) /
NULLIF(SUM(spend)::float, 0)))*100),0))::numeric,2)  AS ROMI
FROM google_ads_basic_daily
GROUP BY ad_date
),
all_ as (
SELECT ad_date, sum(romi) as all_romi
FROM romi_list
GROUP BY ad_date
)
SELECT ad_date, max(all_romi) as daily_max_romi
FROM  all_ group by ad_date
ORDER BY daily_max_romi desc limit 5;
```

2.2 Obtained Results

	ad_date	daily_max_romi
1	2022-01-11	148.69
2	2022-01-07	145.66
3	2022-02-13	141.87
4	2022-01-16	139.79
5	2022-02-23	137.32

The top 5 days with the highest ROMI values were identified.

3. Finding the campaign with the highest total value per week

3.1 SQL Query

```
WITH f_g_united as (
SELECT date_trunc('week', ad_date)::date as week_start, 'facebook' as ad_type,
campaign_name, sum(value) as total_value
FROM facebook_ads_basic_daily fabd
LEFT JOIN facebook_campaign fc on fabd.campaign_id = fc.campaign_id
GROUP BY week_start, campaign_name
UNION ALL
SELECT date_trunc('week', ad_date)::date as week_start, 'google' as ad_type,
campaign_name, sum(value) as total_value
FROM google_ads_basic_daily
GROUP BY week_start, campaign_name
),
all_united as (
SELECT week_start, campaign_name, sum(total_value) as weekly_total_value
FROM f_g_united
GROUP BY week_start, campaign_name
),
max_values_ordered as (
week_start,
campaign_name,
weekly_total_value,
max(weekly_total_value)
over(partition by week_start order by week_start)  as weekly_max_total_value
FROM all_united
)
SELECT
week_start,
campaign_name
```

```
FROM max_values_ordered
WHERE weekly_total_value = weekly_max_total_value
ORDER BY week_start;
```

3.2 Obtained Results

	week_start	campaign_name
1	2020-11-09	Discounts
2	2020-11-16	Discounts
3	2020-11-23	Discounts
4	2020-11-30	Discounts
5	2020-12-07	Discounts
6	2020-12-14	Discounts
7	2020-12-21	Discounts
8	2020-12-28	Discounts
9	2021-01-04	Discounts
10	2021-01-11	Discounts
11	2021-01-18	Discounts
12	2021-01-25	Discounts
13	2021-02-01	Discounts
14	2021-02-08	Discounts
15	2021-02-15	Discounts
16	2021-02-22	Discounts
17	2021-03-01	Brand
18	2021-03-08	Brand
19	2021-04-05	Brand
20	2021-04-12	Electronics
21	2021-04-19	Electronics
22	2021-04-26	Electronics
23	2021-05-03	Electronics
24	2021-05-10	Electronics
25	2021-05-17	Lookalike

The campaign names that generated the most revenue on a weekly basis have been identified.

4. Identifying the campaign with the largest monthly reach increase

4.1 SQL Query

```
WITH campaign_interactions as (
    SELECT to_char((date_trunc('month', ad_date)), 'mm-yyyy') as ad_month,
concat('facebook: ', campaign_name) as campaign_name, max(reach) as max_interaction
    FROM facebook_ads_basic_daily fabd
    LEFT JOIN facebook_campaign fa on fa.campaign_id = fabd.campaign_id
    GROUP BY ad_month, campaign_name
    UNION ALL
    SELECT to_char((date_trunc('month', ad_date)), 'mm-yyyy') as ad_month, concat('google:
', campaign_name) as campaign_name, max(reach) as max_interaction
    FROM google_ads_basic_daily
    GROUP BY ad_month, campaign_name
),
comparison_for_months as (
    SELECT
        ad_month,
        campaign_name,
        max_interaction,
        max_interaction - lag(max_interaction) over (partition by campaign_name order by
ad_month) as interaction_difference
    FROM campaign_interactions
),
top_difference as (
    SELECT
        ad_month,
        campaign_name,
        max_interaction,
        interaction_difference,
        max(interaction_difference) over (partition by campaign_name) as top_difference
```

```
    FROM comparison_for_months
),
campaings_for_all_months as (
    SELECT ad_month, campaign_name, interaction_difference
    FROM top_difference
    WHERE interaction_difference = top_difference
),
best_campaigns_for_all_months as (
    SELECT
        ad_month,
        campaign_name,
        interaction_difference,
        max(interaction_difference) over (partition by ad_month) as best_difference
    FROM campaings_for_all_months
)
SELECT ad_month, campaign_name as best_campaign
FROM best_campaigns_for_all_months
WHERE interaction_difference = best_difference
ORDER BY ad_month;
```

4.2 Obtained Results

	ad_month	best_campaign
1	04-2022	google: Hobbies
2	05-2021	facebook: Lookalike
3	06-2021	google: Wholesale
4	07-2021	facebook: Electronics
5	11-2020	facebook: Discounts
6	12-2020	facebook: Brand
7	12-2021	facebook: Trendy

This query performed a "Campaign Momentum Analysis." Instead of just looking at total reach, it identified the "monthly champion" campaigns that increased their reach the most (made the biggest jump) compared to the previous month. This is an advanced marketing analytics method used to see which campaigns have reached saturation and which are rapidly spreading to new target audiences.

5. Writing a query that shows the adset_name and duration with the longest continuous impression

5.1 SQL Query

```
WITH adset_days as (
    SELECT concat('facebook: ', adset_name) as adset, ad_date
    FROM facebook_ads_basic_daily fabd
    LEFT JOIN facebook_adset fa on fa.adset_id = fabd.adset_id
    UNION ALL
    SELECT concat('google: ', adset_name) as adset, ad_date
    FROM google_ads_basic_daily
),
adset_ordered as (
    SELECT adset, ad_date
    FROM adset_days
    ORDER BY adset, ad_date
),
numbered_days as (
    SELECT
        adset,
        ad_date,
        row_number() over (partition by adset order by ad_date) as row_num
    FROM adset_ordered
    WHERE ad_date is not null
),
continuous_periods as (
    SELECT
```

```
        adset,
        ad_date,
        ad_date - INTERVAL '1 day' * row_num AS period_id
        -- By subtracting the row number in days from the current date,
        -- we get a constant date for consecutive days. If the sequence breaks,
        -- this result changes, allowing us to use it to identify a single period.
    FROM numbered_days
),
period_durations AS (
    SELECT
        adset,
        period_id,
        min(ad_date) AS start_date,
        max(ad_date) AS end_date,
        count(*) AS consecutive_days
    FROM continuous_periods
    GROUP BY adset, period_id
    ORDER BY adset
)
SELECT
        adset,
        max(consecutive_days) as max_consecutive_days
FROM period_durations
GROUP BY adset
ORDER BY max_consecutive_days desc
limit 1;
```

5.2 Obtained Results

	adset	max_consecutive_days
1	facebook: Wide	50

To identify the maximum continuous performance of ad sets across Google and Facebook, I implemented a 'Gaps and Islands' analysis using SQL window functions. By calculating the difference between sequential calendar dates and a partitioned row number, I generated a unique 'Period ID' for each uninterrupted run of impressions. This mathematical approach allowed me to group consecutive active days into distinct 'islands' of activity, enabling a precise measurement of the longest streak (Maximum Consecutive Days) for each campaign. This analysis provides critical insights into the operational stability and delivery consistency of our advertising assets.